

EDUVISION

Higher Education Intelligence System

Institutional Performance and Academic Excellence Analysis

PROJECT DOCUMENTATION

Project focus: higher-education performance analysis through data preparation, KPI engineering, and four Tableau dashboards.

Dashboard suite

Dashboard	Purpose
University Overview	Institution-level summary, ranking and regional patterns.
Research Analytics	Research productivity and impact comparisons.
Student Analytics	Enrollment, international participation and faculty/student measures.
Country Comparison	Country- and region-level comparisons.

Prepared as the final documentation report for the EduVision project.

Document control and scope

Purpose

This report explains the EduVision project from data workflow through dashboard design, analysis, validation, and delivery. It is intended for mentors, evaluators, teammates, and portfolio reviewers.

Evidence and scope

The report uses project context available in this conversation: milestone names, field names, dashboard descriptions, visible quality issues from screenshots, and the provided live Tableau URL. The GitHub README could not be fetched during preparation, so exact dataset citations, licence details, and some implementation details must be confirmed from the repository before final submission.

Live dashboard

<https://prod-in-a.online.tableau.com/t/punikotiharsha-4b92357949/views/EduVision/UniversityOverview/5ac0bf9b-feb2-46e8-880f-48e557ac455b/1290284c-05e2-4bb8-95a9-80ca4dc1c5f2>

GitHub branch

<https://github.com/springboardmentor1234r/Higher-Education-Intelligence-System-for-Institutional-Performance-and-Academic-Excellence-Analysis/tree/PUNIKOTI-HARSHA>

Contents

- Project overview and objectives.
- Dataset, collection, cleaning, and KPI engineering.
- Four dashboard guides and filter architecture.
- Testing, limitations, deployment, and final delivery.

1. Executive summary

EduVision is a higher-education analytics project designed to bring institution, research, student, and country-level indicators into one Tableau dashboard suite. The goal is to make performance indicators easier to explore than disconnected tables and charts.

Four analytical views form the dashboard suite. University Overview gives the broad performance picture; Research Analytics focuses on productivity and impact; Student Analytics covers enrollment and international participation; Country Comparison supports cross-country and regional review.

Main outputs

- Prepared university dataset used by Tableau.
- Data collection, cleaning, and KPI engineering scripts where present in the repository.
- Four Tableau dashboard pages and supporting worksheets.
- Live Tableau link for interactive review.
- Documentation of sources, KPIs, usage, methodology, and delivery.

Validation status

The supplied screenshots show items to verify before final evaluation: null values on some charts, one unknown geographic mark, possible year formatting issues, and chart labels or scales needing review. These are recorded as validation tasks, not claimed fixes.

2. Problem context and motivation

Higher-education performance is multi-dimensional. Institutions may be compared using ranking, academic reputation, research impact, international participation, student enrollment, and faculty capacity. Reviewing each indicator separately can make comparison difficult.

EduVision addresses this by presenting selected measures in interactive charts with shared filters. Viewers can start with the overall university picture, inspect research and student patterns, then compare countries or regions.

Intended users

- Students and evaluators reviewing institutional performance patterns.
- Academic analysts exploring research and student indicators.
- Mentors or stakeholders reviewing the workflow and dashboard outputs.
- Portfolio reviewers assessing the data-to-dashboard implementation.

Questions supported

- Which institutions have higher values for a selected ranking or performance measure?
- How do ranking-related indicators vary across years and regions?
- How do research productivity and impact compare?
- How do international-student percentages and faculty/student indicators vary?
- How do selected measures differ across countries?

Interpretation boundary

The dashboards support descriptive comparison. They do not by themselves establish causation or prove that an institution is better in every dimension. Conclusions should be limited to the actual data, years, filters, and fields represented.

3. Objectives and scope

Objectives

- Prepare university data for analysis.
- Apply consistent cleaning and transformation rules.
- Create reusable education-related KPIs from available fields.
- Develop four complementary Tableau dashboards.
- Provide filters and navigation for exploration.
- Document methodology, KPI meaning, data limitations, and usage.
- Deliver source artifacts through GitHub and a Tableau link.

In scope

The described scope includes data preparation, descriptive KPI engineering, worksheet development, dashboard assembly, filter behavior, visual QA, workbook packaging, documentation, and delivery links.

Not assumed to be in scope

- Predictive modelling or causal inference not shown in the implementation.
- Automated refresh infrastructure not confirmed by the available project context.
- Claims that every calculation or interaction has passed testing without evidence.
- Claims that the dataset is official or complete without source citations.

Success criteria

Area	Evidence
Data	Sources, grain, fields, missingness and transformations documented.
KPIs	Calculation, aggregation, units and interpretation documented.
Dashboards	Four pages load and support their intended analysis.
Interactions	Filters and navigation tested on relevant worksheets.
Delivery	Workbook and documentation accessible through intended links.

4. Architecture and milestones

The project is organized into milestone-based work. Earlier milestones prepare data and engineer KPIs; later milestones develop, test, and deliver the dashboards.

Milestone	Purpose	Artifacts referenced
1. Data Collection and Preparation	Collect and prepare university records.	university_raw_data.csv; data_collection.py; university_cleaned.csv; education_cleaning.ipynb
2. KPI Engineering and Dashboard Planning	Generate analysis-ready fields and plan dashboards.	university_final_dataset.xlsx; generate_education_kpis.py; storyboard/prototype
3. Dashboard Development	Build four Tableau dashboards.	Tableau workbook and dashboard pages
4. Testing and Delivery	Validate outputs and document project.	QA checklist, test report, final documentation, workbook links

Logical workflow

Stage	Input	Output
Collect	Source records	Raw dataset
Clean	Raw records and rules	Cleaned dataset
Engineer	Cleaned fields	Final KPI dataset
Visualize	Final dataset	Worksheets and dashboards
Validate	Data and workbook	Test evidence and defect list
Deliver	Validated artifacts	Repository, report and live link

5. Dataset overview and grain

The dashboard context indicates a university-level analytical dataset with institution identifiers and descriptive dimensions alongside ranking, research, and student-related measures. Exact source, licence, row count, and coverage period must be recorded from the actual source and README.

Expected grain

Many views appear to use a university record associated with a country, region, and year. Confirm whether there is one record per university-year or whether multiple source records exist. Duplicate records can distort counts and averages.

Field	Role	Usage
university_name	Dimension	Institution labels, ranking and tooltips.
university_id	Identifier, if present	Stable institution identity.
country	Dimension	Country comparison and maps.
region	Dimension	Regional grouping.
year	Time dimension	Filters and trends.
global_ranking_score	Measure	Ranking/performance comparisons.
academic_reputation_score	Measure	Reputation comparisons.
research_impact_score	Measure	Research impact.
research_productivity_index	Measure	Productivity analysis.
international_student_pct	Measure	International participation.
faculty_student_ratio	Measure	Faculty/student indicator.

6. Dataset sources and governance

The final report must identify the original dataset source, not only the cleaned workbook. Record the publisher, dataset title, source URL, access date, coverage years, and licence. The project README could not be fetched during preparation, so the source URL and licence are not invented here.

Required source register

Item	What to record
Source name	Organization, publisher or dataset owner.
Dataset title	Exact title from the source.
URL / citation	Stable URL or formal citation.
Access date	Date accessed or downloaded.
Coverage	Years, institutions, geography and subjects.
Licence	Reuse, redistribution and attribution conditions.
Transformation	Conversions, joins, filters and calculated fields.

Data handling principles

- Preserve raw data when permitted and separate it from cleaned outputs.
- Record transformations for reproducibility.
- Do not replace missing values with zero unless justified.
- Do not publish restricted or personal data.
- Keep source and licence notes in the repository.

Before submission

Complete this section from the actual README and source files. Do not submit guessed publisher names, record counts, or licence terms as facts.

7. Data collection and ingestion

Data collection establishes the raw records used by later cleaning and analysis. Project context identifies a collection script and raw university CSV as Milestone 1 artifacts. The exact provider and collection mechanism must be confirmed from the script and README.

Ingestion record

- Record the source URL or file origin and collection date.
- Keep a raw copy unchanged when permitted.
- Inspect encoding, delimiter, headers, and data types.
- Check institution names and identifiers across years.
- Record row count, column count, duplicate keys, and missingness.

Initial profiling checklist

Check	Question
Schema	Are expected columns present and consistent?
Types	Are year, scores, percentages and ratios suitable types?
Uniqueness	Does the selected university/year key identify a row?
Missingness	Which fields have null or blank values?
Range	Are numeric values plausible under documented definitions?
Categories	Are country and region labels standardized?

Output

The raw dataset and collection script form the starting point for cleaning. Source-specific collection assumptions should be documented in the repository README.

8. Cleaning and transformation

Cleaning improves consistency while preserving source meaning. Project context identifies a cleaned CSV and a cleaning notebook in Milestone 1. The exact transformation rules should be verified against those artifacts.

Core cleaning tasks

- Normalize column names and remove accidental whitespace.
- Standardize institution, country and region labels carefully.
- Convert year fields to a consistent representation.
- Convert scores, percentages and ratios to numeric types.
- Check duplicates at the intended university/year grain.
- Review nulls and outliers and document treatment.
- Keep display strings separate from numeric measures.

Missing values and outliers

A null means a value is absent or unavailable; it is not automatically zero. Outliers should be reviewed against source and domain context. Do not remove a valid record only because it changes a chart scale.

Transformation log template

Issue	Rule applied	Reason	Evidence
Year	Record conversion/format rule	Consistent time axis	Type check
Country/region	Record standardized labels	Reliable geography	Category audit
Numeric KPI	Record parsing/null rules	Correct aggregation	Range/null summary
Duplicates	Record key and resolution	Avoid double-counting	Duplicate report

9. KPI engineering and definitions

KPI engineering converts prepared fields into indicators that answer analytical questions. Definitions below express intended concepts for fields referenced in project context. Verify the actual Tableau aggregation and formula before treating them as final.

KPI / measure	Conceptual definition	Caution
Total Universities	Distinct count of university IDs or names in current filters.	Prefer stable IDs when available.
Average Overall Score	Average of the documented overall-score field.	Confirm field and formula exist.
Global Ranking Score	Ranking-related score for institution comparison.	Confirm whether larger or smaller means better.
Academic Reputation Score	Reputation measure at the record level or aggregated.	Confirm scale and aggregation.
Research Impact Score	Measure representing research impact.	Do not infer citation meaning without definition.
Research Productivity Index	Index used to compare productivity.	Do not call it publication count without evidence.
International Student %	Percentage as defined by source or KPI script.	Check whether stored 0–1 or 0–100.
Faculty-to-Student Ratio	Ratio as defined by source.	Confirm direction and numeric aggregation.

10. KPI calculations and formatting

A KPI requires a calculation and an explicit interpretation. Label, number format, aggregation, and filter context should agree.

Calculation principles

- Use COUNTD on a stable institution identifier for unique university count where available.
- Use AVG only when analytically meaningful and supported by the source field.
- Document whether a measure is a score, count, ratio, percentage, or index.
- Keep numeric fields numeric and use separate display fields for formatted strings.
- Ensure chart titles match the plotted field.
- Confirm the direction of ranking measures before sorting.

Percentage formatting

Determine the storage convention first. If values are fractions such as 0.25, percentage formatting displays 25%. If values are already stored as 25, percentage formatting may display 2,500%. Inspect source values and calculations before changing formats.

Faculty ratio formatting

If faculty_student_ratio_display contains text such as “1 : X”, use it for labels only. Use the numeric faculty_student_ratio field for sorting, averaging, and axes.

KPI validation

Validation	Check
University count	Compare Tableau COUNTD with distinct-key count.
Average score	Recalculate a sample and confirm filter scope.
Percentage	Check storage scale, axis and label formatting.
Ratio	Confirm numerator/denominator and aggregation.
Ranking	Confirm which direction indicates better performance.

11. Tableau workbook and design

The Tableau workbook connects the prepared dataset to worksheets and dashboard pages. The project context identifies four dashboards and a packaged workbook (.twbx) for sharing. A .twbx can package workbook data and supporting assets; a .twb is a workbook definition and may depend on external connections.

Design principles

- Consistent dark navy styling with purple headings and KPI cards.
- Common Year, Region and Country filters.
- Navigation between the four dashboard pages.
- Bars for rankings, lines for time, scatter plots for relationships, maps for geography, and tables for detail.
- Tooltips that clarify institution, country, year and measure.

Workbook organization

The workbook is expected to contain supporting worksheets assembled into four dashboard pages. Exact worksheet names and calculations should be confirmed in Tableau Desktop; screenshots show only part of the workbook structure.

External extension

The Country Comparison screenshot shows an external radar/spider visualization associated with Infotopics. Document this dependency and confirm availability in the target environment. Consider a native Tableau comparison view as a fallback if portability is important.

12. Dashboard guide — University Overview

Purpose

University Overview is the entry point for institution-level performance. It combines headline KPIs with rankings, regional distribution and trend views.

Visible components

- Top Global Ranking indicator.
- KPI cards such as Global Universities, Avg Overall Score, International Students %, Faculty Ratio and Research Productivity Index.
- Top 10 Universities by Global Ranking Score bar chart.
- Average Global Ranking Score by Year and Region line chart.
- Universities by Region donut chart.
- Top universities by Research Productivity Index.
- International Students % by Region map.
- Students per Faculty by Region bar chart.

How to use

- Read KPI cards under the current filter context.
- Use ranked bars to compare institutions.
- Use the trend chart to review changes across available years.
- Use regional views for distribution and faculty/student patterns.
- Confirm filters are configured for every intended worksheet.

QA note

The screenshot context showed nulls on the ranking trend and a year displayed as “2,005”. Verify the Year data type/number format and investigate missing ranking values. Ranking and score measures are not automatically interchangeable.

13. Dashboard guide — Research Analytics

Purpose

Research Analytics focuses on productivity, impact, institution-level comparisons and year-based patterns.

Visible components

- Top 10 Publications/productivity ranking view.
- Impact versus Productivity scatter plot.
- Research benchmark table with institution/country and research-related measures.
- Research Productivity Trend by year.
- Citation Top 10 comparison.
- Common filters and dashboard navigation.

How to use

- Use ranking bars to compare the displayed research measure.
- Use the scatter plot to inspect how productivity and impact vary together; correlation is not causation.
- Use the trend chart to compare productivity across years.
- Use the table to inspect institution-level values.

QA note

Screenshots showed null counts on the scatter and trend charts. Some titles refer to publications while values appear decimal. Verify the field, units, aggregation, and missing-value handling; align chart titles and axes with the actual calculation.

14. Dashboard guide — Student Analytics

Purpose

Student Analytics summarizes student population, enrollment, international participation and faculty/student capacity measures.

Visible components

- Student and enrollment KPI cards, including estimated total students, average enrollment, university count and domestic enrollment as shown.
- Student distribution treemap.
- International-student percentage by university.
- Enrollment comparison over time.
- Student-diversity trend.
- Students-per-faculty comparison by region.

How to use

- Read KPI cards under current filters.
- Use treemap size to inspect the represented distribution.
- Compare international-student percentages after confirming scale.
- Use trends to review changes over time.
- Confirm the ratio direction before interpreting regional bars.

QA note

Screenshots showed null indicators in the treemap and diversity trend. Percentage labels and axis scale may not align. Confirm whether values are stored as fractions or 0–100 and validate both labels and axis.

15. Dashboard guide — Country Comparison

Purpose

Country Comparison shifts analysis from individual institutions to country- and regional-level comparisons.

Visible components

- Country Ranking Table.
- Country Radar Comparison visualization.
- International Students % by Region map.
- Research Impact Score by Country bar chart.
- Ranking-score trend by year and region.

How to use

- Use the table to inspect country-level values.
- Use the radar view only with intended measures and comparable scales.
- Use the map for geographic context after resolving unknown locations.
- Use bars to compare country research impact.
- Use filters to narrow the year or region where applicable.

QA note

The screenshot context showed one unknown map location and an external radar extension. Review geographic roles and spelling, confirm the measures in the radar chart, and document the extension dependency.

16. Filters and interaction design

The four pages are intended to share Year, Region and Country filters and navigation buttons. A filter on a dashboard does not automatically affect every worksheet; its scope must be configured and tested.

Filter checklist

- Use Apply to Worksheets or select intended worksheets in the filter menu.
- Confirm worksheets use compatible data sources and matching fields.
- Test cascading Region-to-Country behavior if intended.
- Test filters individually and in combination.
- Verify reset behavior and make active filter state visible.

Navigation checklist

- Each navigation button opens the correct dashboard.
- Filter placement and visual styling remain consistent.
- No button points to a missing or outdated view.
- The published URL opens the intended workbook and view.

Dashboard actions

If a chart should filter or highlight other charts, configure the dashboard action explicitly and test it. Record whether clicking a mark filters, highlights, navigates, or only selects. Do not claim an interaction works until verified.

17. Testing and quality assurance

Testing must verify data correctness and the viewer experience. Complete the checklist with actual outcomes before claiming full validation.

Area	Test	Evidence to record
Data	Schema, duplicates, missing values, ranges	Pass/fail, issue ID
KPI	Compare KPI with independent calculation	Expected/actual values
Filters	Year, Region, Country across relevant sheets	Sheets affected and result
Navigation	Visit all four dashboards	Pass/fail and notes
Visuals	Titles, units, formats, labels, tooltips	Observed defects
Map	Resolve unknown geographic values	Value and resolution
Extension	Open radar in target environment	Dependency/version
Packaging	Reopen .twbx	Data and visuals available

Known items to verify

- Null values on ranking, research or student charts.
- Year formatting that appears as "2,005".
- Research titles and decimal values.
- International-student percentage axis and label scaling.
- One unknown map location.
- Radar measures and extension dependency.

Issue management

Record an issue ID, dashboard/worksheet, reproduction steps, expected and actual results, severity, fix, and retest outcome. Keep unresolved issues visible in the final report.

18. Delivery, limitations and conclusion

Recommended repository deliverables

- README.md with overview, setup instructions, dashboard links and folder map.
- Dataset source register and licence notes.
- KPI definitions and education analytics methodology.
- Dashboard user guide and final project documentation PDF.
- Tableau workbook (.twb or .twbx) and extension notes.
- Testing checklist and testing report with actual outcomes.
- Scripts and datasets only where redistribution is permitted.

Live Tableau dashboard

<https://prod-in-a.online.tableau.com/t/punikotiharsha-4b92357949/views/EduVision/UniversityOverview/5ac0bf9b-feb2-46e8-880f-48e557ac455b/1290284c-05e2-4bb8-95a9-80ca4dc1c5f2>

GitHub branch

<https://github.com/springboardmentor1234r/Higher-Education-Intelligence-System-for-Institutional-Performance-and-Academic-Excellence-Analysis/tree/PUNIKOTI-HARSHA>

Limitations

This report does not independently verify the exact dataset provenance, every workbook calculation, or live dashboard permissions. The repository README could not be fetched during preparation. Confirm source citations, licence terms, KPI formulas, and final branch state. QA items are checks to perform, not claims that defects are resolved.

Conclusion

EduVision combines university overview, research analytics, student analytics, and country comparison in a Tableau-oriented project. Final quality depends on clear source documentation, defensible KPI definitions, correct formatting, tested filters/navigation, and reproducible delivery. Completing the QA checklist and confirming README-specific details will strengthen the final submission.